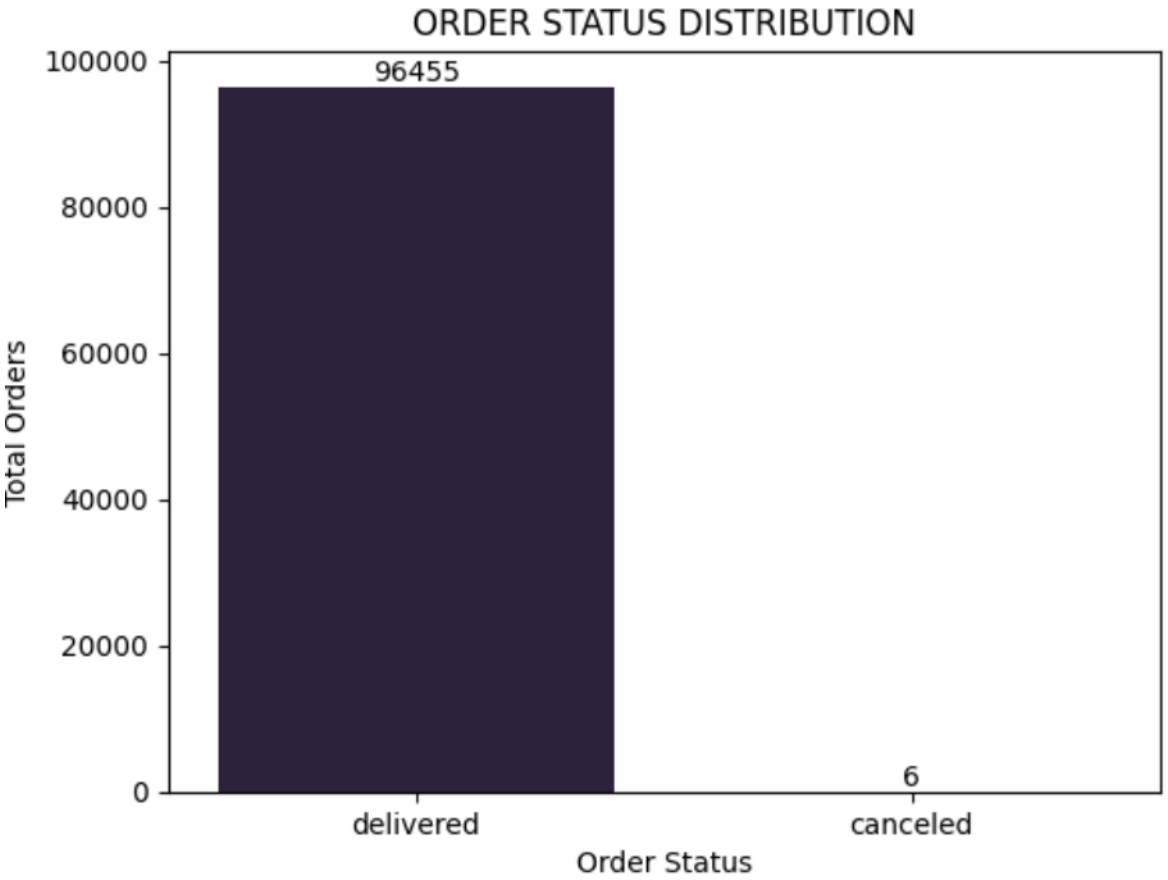
Customer Purchase Behavior Analysis in E-Commerce

VISUALIZATIONS:

1.ORDER STATUS DISTRIBUTION:

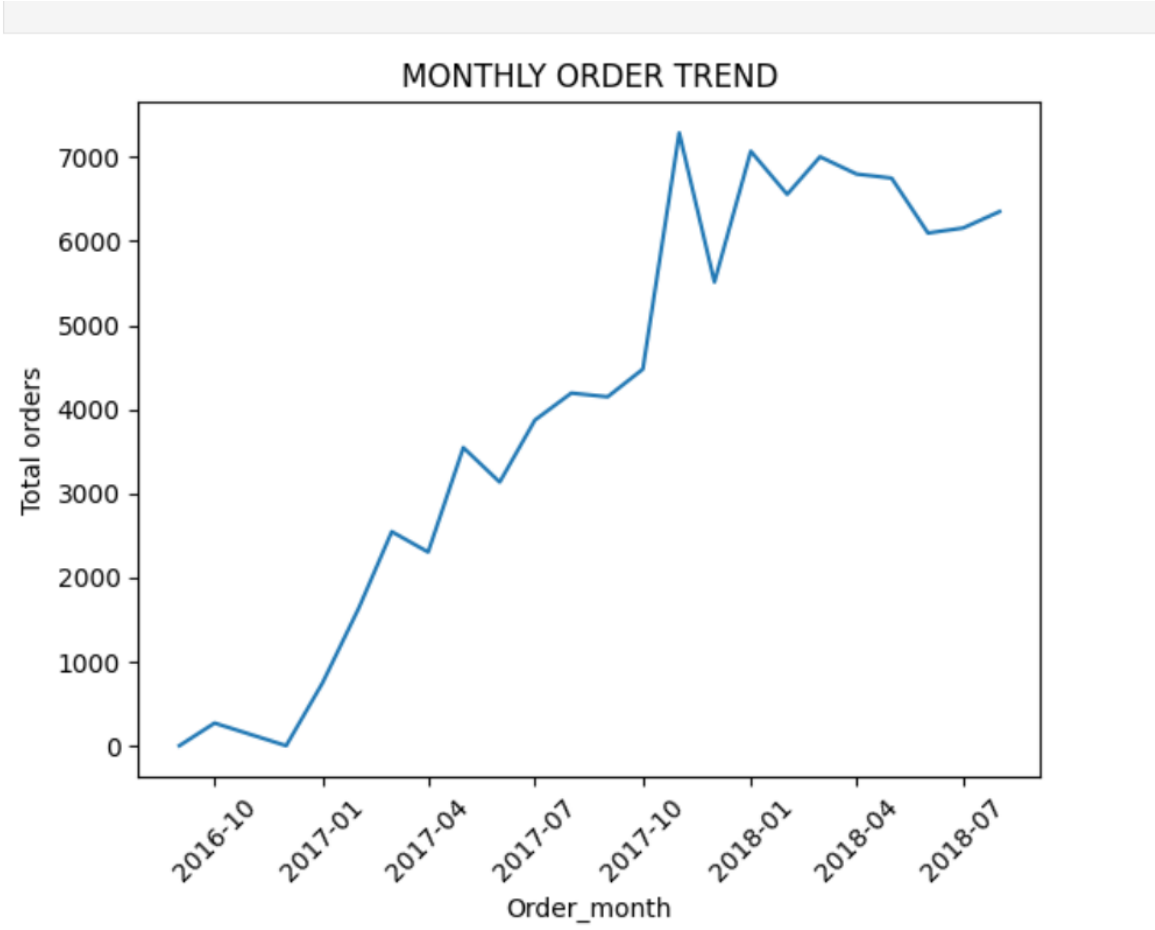


KEY INSIGHT:

Most orders fall under the “delivered” status, indicating that the platform successfully completes the majority of transactions. A

smaller portion of orders are either canceled or unavailable, suggesting potential operational or inventory issues. Monitoring these unsuccessful orders can help improve fulfillment reliability.

2.MONTHLY ORDER TREND:

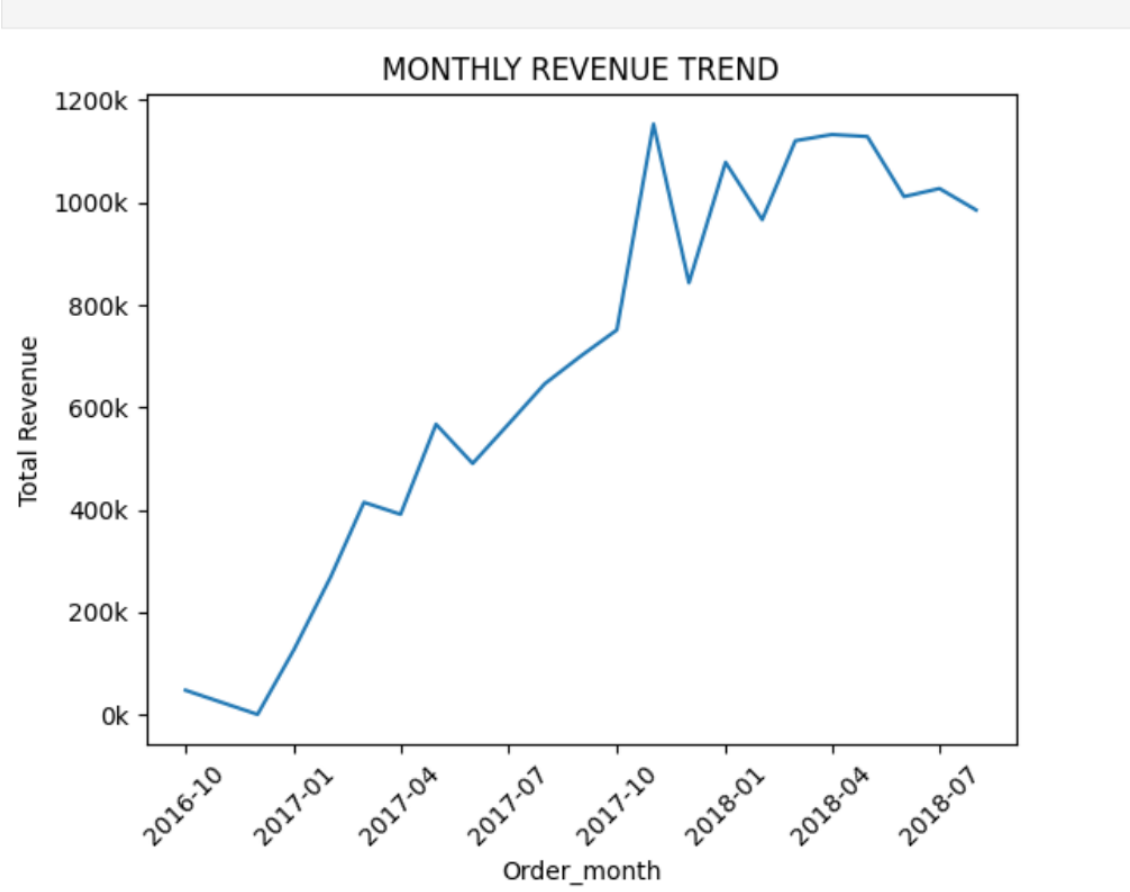


KEY INSIGHT:

Order volume shows clear fluctuations over time, with certain months experiencing higher activity. These peaks likely correspond to promotional events or seasonal demand. Understanding these

patterns allows the business to better plan inventory levels and marketing campaigns.

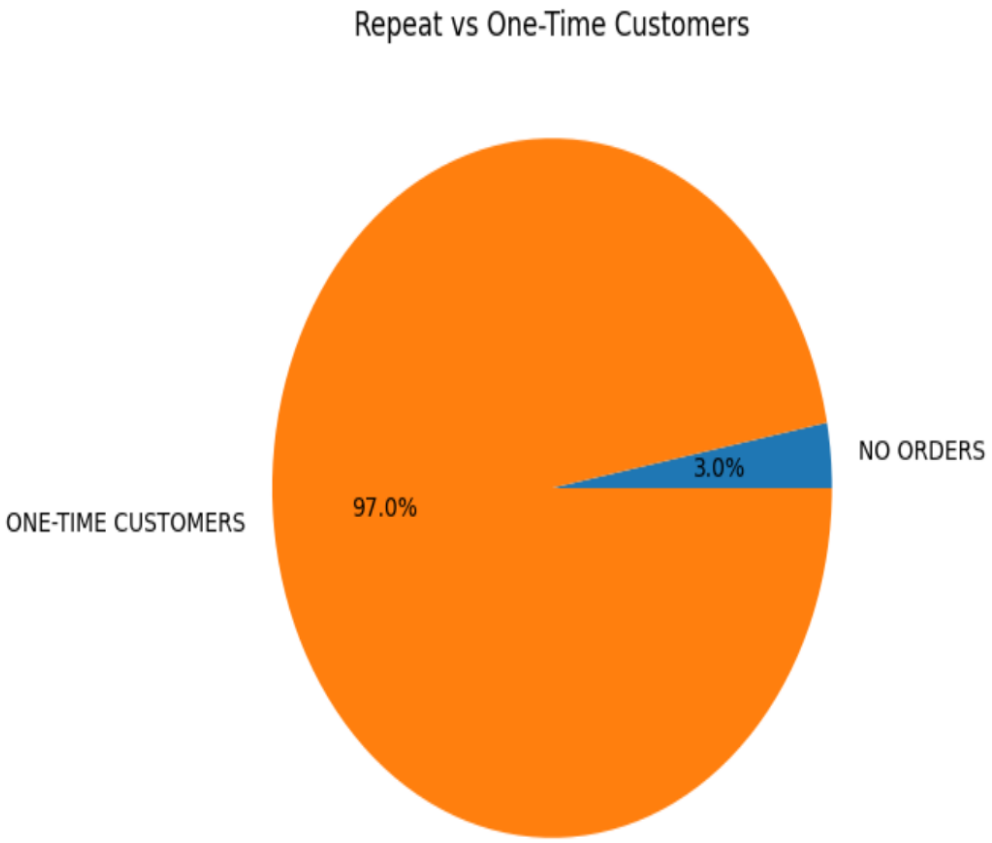
3.MONTHLY REVENUE TREND:



KEY INSIGHT:

Revenue generally follows the same trend as order volume, confirming that higher order activity directly drives revenue growth. However, some months generate higher revenue without a significant increase in orders, suggesting that higher-value purchases may occur during those periods.

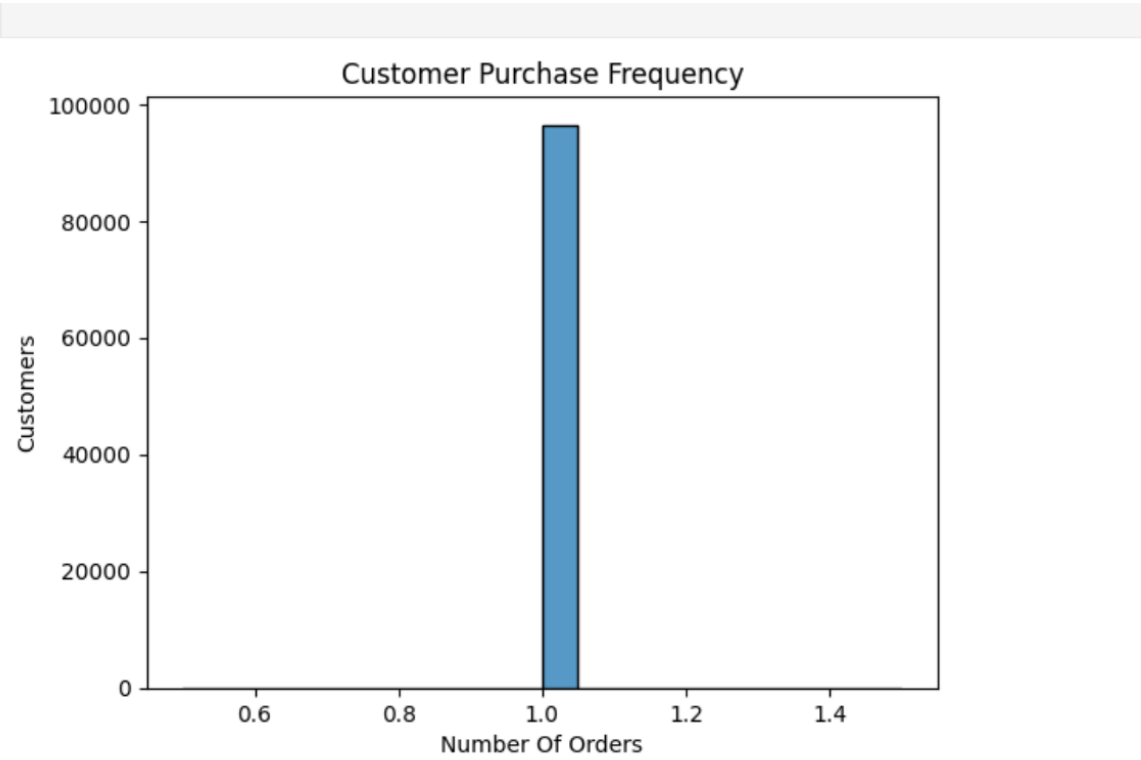
4.REPEAT VS ONE-TIME CUSTOMERS:



KEY INSIGHT:

A significant portion of customers make only one purchase, indicating an opportunity to improve customer retention. Increasing repeat purchases through loyalty programs, targeted promotions, or personalized recommendations could significantly boost overall revenue.

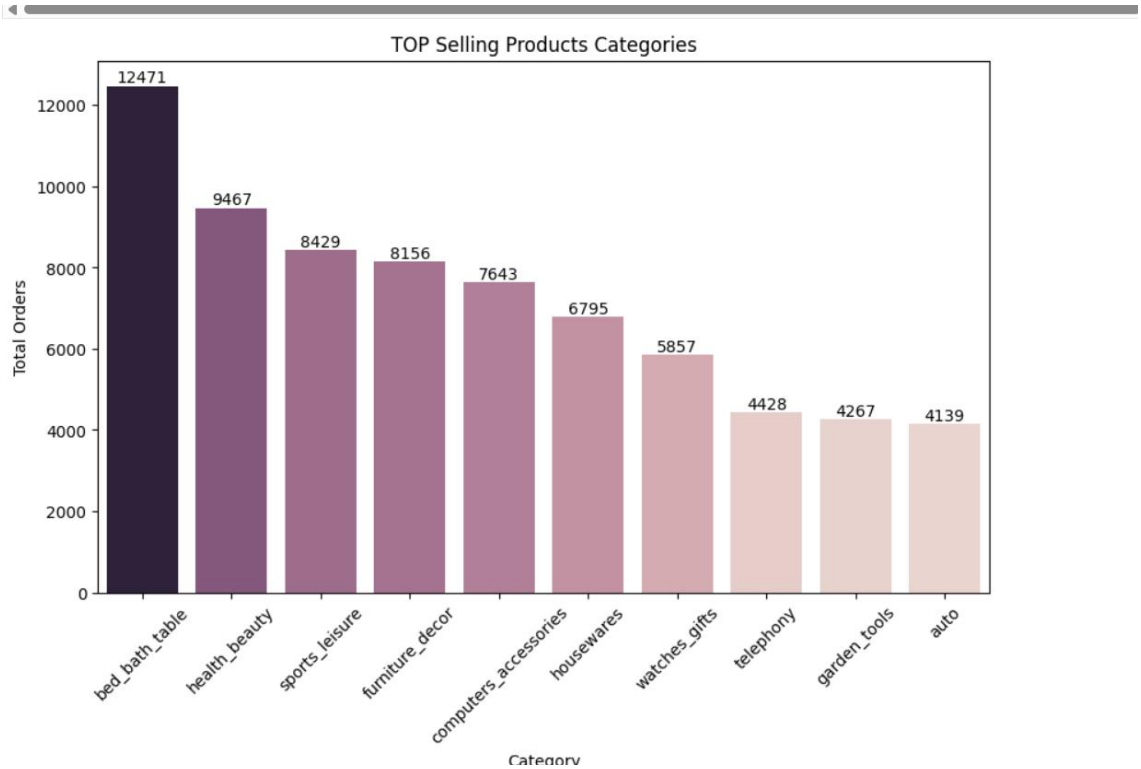
5.CUSTOMER PURCHASE FREQUENCY:



KEY INSIGHT:

Most customers place only a small number of orders, while a smaller group of customers purchases frequently. These frequent buyers represent high-value customers, and understanding their behavior could help the business design targeted retention and engagement strategies.

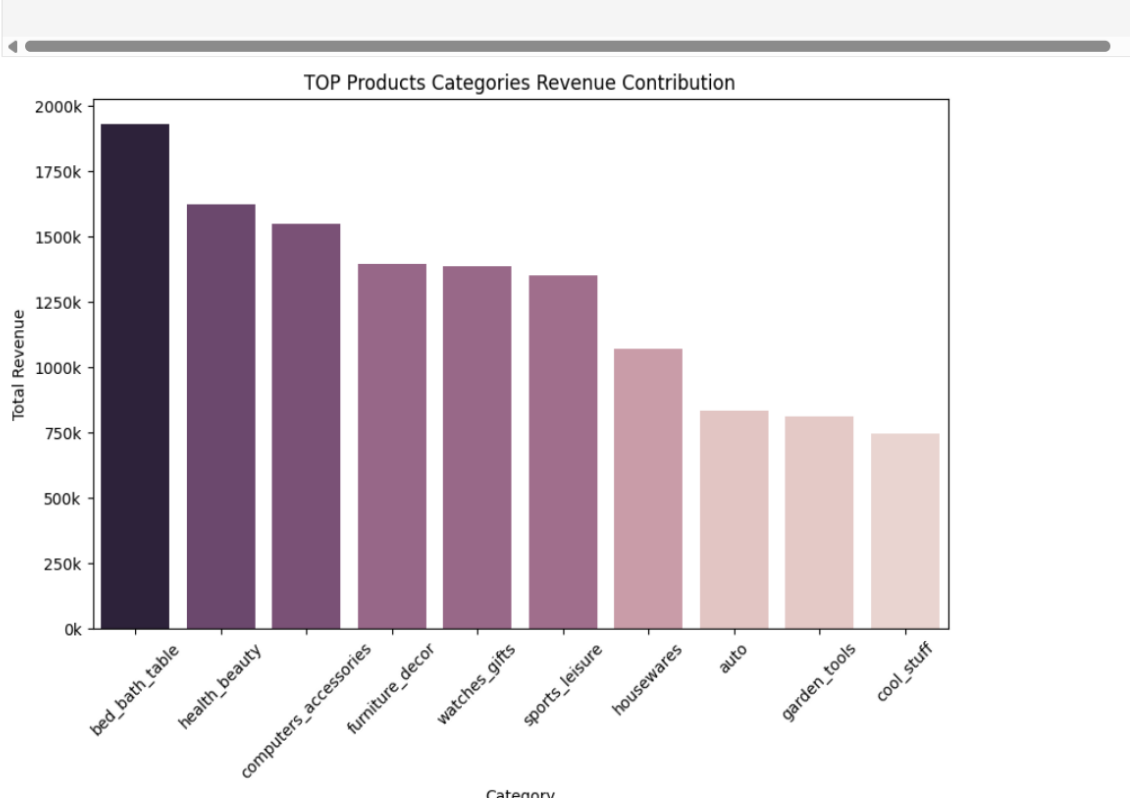
6.TOP PRODUCT CATEGORIES BY SALES:



KEY INSIGHT:

A limited number of product categories account for the majority of sales. This indicates that customer demand is concentrated around specific types of products. These high-performing categories should receive priority in marketing campaigns and inventory planning.

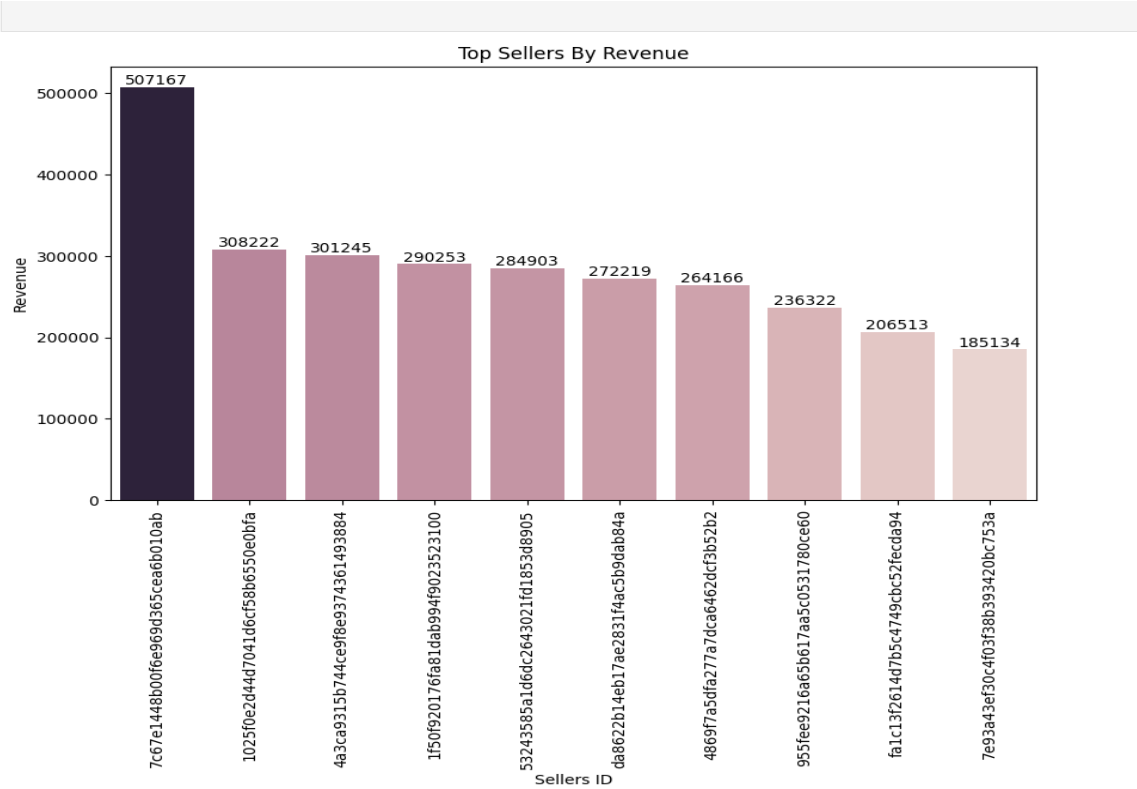
7.REVENUE BY PRODUCT CATEGORY:



KEY INSIGHT:

Some product categories generate higher revenue despite fewer transactions, indicating higher-priced products or premium segments. These categories represent valuable revenue drivers and may benefit from targeted promotion to maximize profitability.

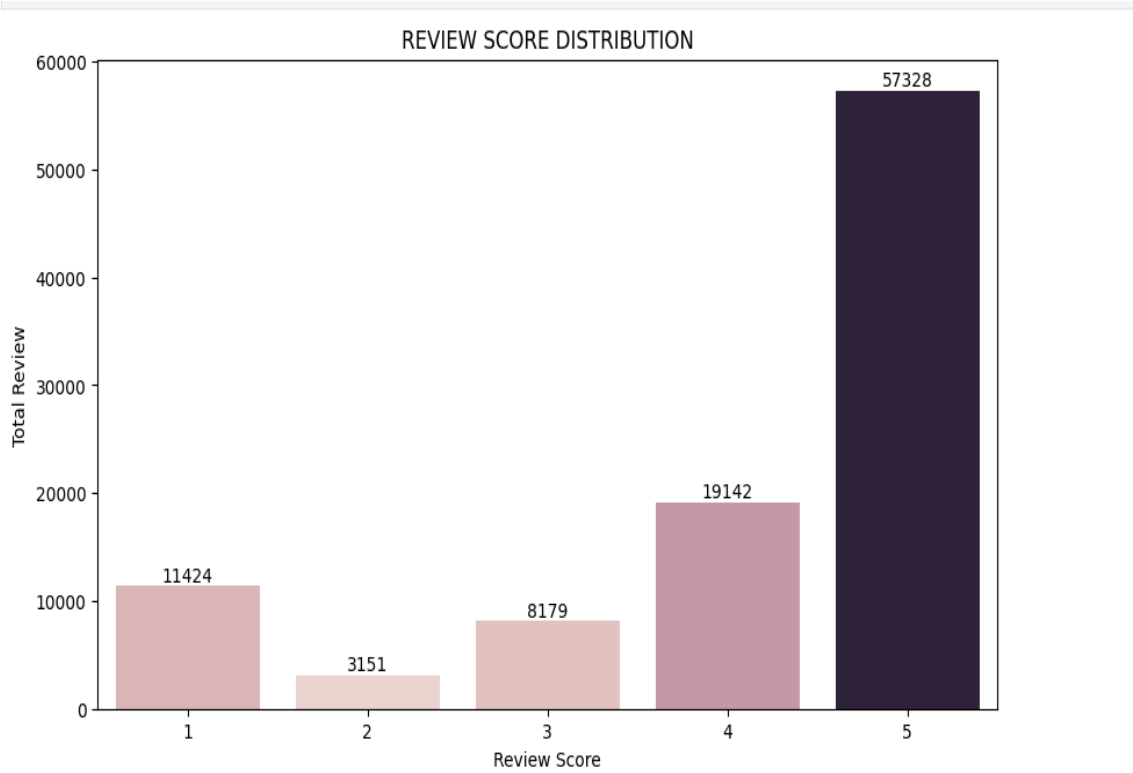
8.TOP SELLERS BY REVENUE:



KEY INSIGHT:

A small number of sellers contribute a disproportionately large share of marketplace revenue. This indicates that the platform relies heavily on high-performing sellers. Supporting smaller sellers while maintaining strong relationships with top sellers could help stabilize marketplace performance.

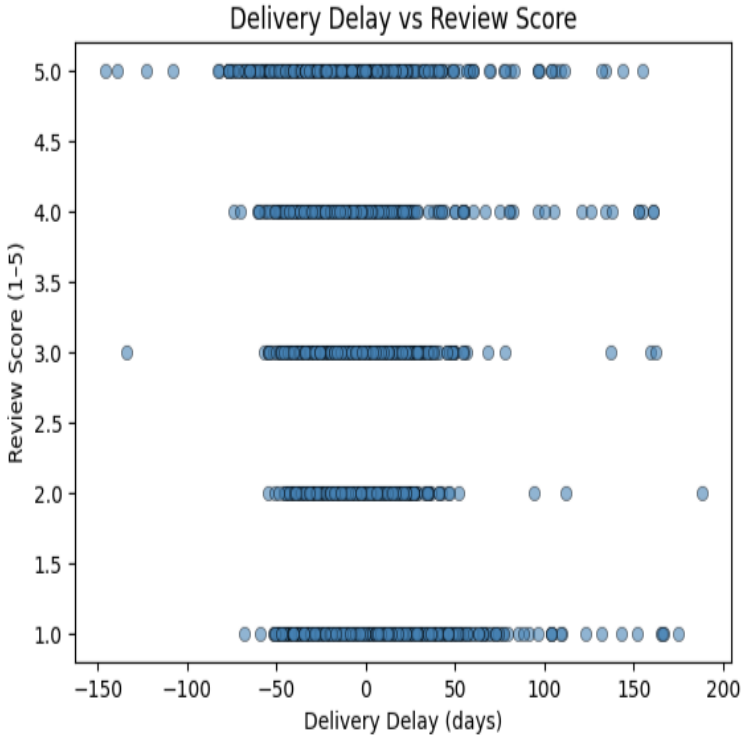
9.REVIEW SCORE DISTRIBUTION:



KEY INSIGHT:

Most reviews fall within the 4–5 star range, suggesting that customers generally have a positive experience on the platform. However, lower ratings still exist and may highlight areas where service quality or product reliability could be improved.

10.DELIVERY DELAY VS REVIEW SCORE:



KEY INSIGHT:

There is a noticeable relationship between delivery delays and lower review scores. Orders that take longer to reach customers tend to receive poorer ratings. This highlights the importance of efficient logistics and reliable delivery services in maintaining customer satisfaction.